FAR AWAY · 2026

OrbitGuard — Deep-Dive Build + Feasibility Report

Space & Aerospace · FAR AWAY 2026

End-to-end pipeline, step-by-step plan, real data sources, honest metrics, and what can break

KEY FEASIBILITY ANSWER

Data sources: LIVE free APIs (CelesTrak + Space-Track CDMs) AND a static official answer key (TraCSS) to prove accuracy — use both.

The engineering: coarse pre-filter sieve (avoids $O(N^2)$ blow-up) + Skyfield/SGP4 propagation + LLM explainer + CesiumJS 3D globe.

Validate screener against the TraCSS answer key — that gives you a real, defensible recall metric.

Prepared by Claude (Opus 4.8) · 11 June 2026 · Deeply researched against the live web

OrbitGuard — project snapshot

What it is	Data sources	Honest metric	Top risk	Rating
Explainable collision screening + 3D globe for small operators	Live CelesTrak/Space-Track + TraCSS key	Recall vs official TraCSS key	Orbital math; Pc needs covariance	**8.2**

The deep-dive below covers: the key feasibility question answered, the end-to-end pipeline, a step-by-step build plan, the realistic result you can claim (with citations), and an honest 'what can break' list.

OrbitGuard

Free, explainable satellite collision-risk screening + 3D globe, validated on the official TraCSS answer key

CLAUDE'S RATING

8.2 / 10

Ingest public orbital data, screen for close approaches, score the risk WITH an explanation, and show a 3D globe with a DODGE / WAIT / WATCH call for small operators.

YOUR QUESTION - live free API or static data? ANSWER: both, and that is the strength
LIVE free data exists: CelesTrak GP serves TLE / OMM element sets, refreshed every 2 hours (cache it - over 100 MB per day or more than one download per update can get your IP firewalled). Space-Track.org (free account) adds TLEs AND real Conjunction Data Messages (CDMs) that carry covariance. STATIC validation exists too: the Office of Space Commerce TraCSS 'Dataset for Conjunction Assessment Verification' ships TWO answer keys (a spherical screening volume and the SFSH rectangular volumes) plus a User's Guide - request access with a Google account.
APPROACH: drive the live 3D globe and screening from CelesTrak TLEs, then VALIDATE your screener against the TraCSS answer key for credible accuracy; use CDM covariance for a real collision probability, or a transparent simplified estimate where it is missing.

WHAT YOU ARE BUILDING

A free, browser-based dashboard that ingests public orbital data, screens for close approaches, scores the risk with an explanation, and shows a 3D globe with a clear DODGE / WAIT / WATCH call for small operators.

END-TO-END PIPELINE (start to finish)

1. Fetch and cache public element sets (CelesTrak GP TLE / OMM; Space-Track for CDMs)
2. Propagate orbits with Skyfield / SGP4 (pure Python)
3. Coarse pre-filter candidate pairs (altitude bands, orbital-geometry sieve) - avoids the O(N squared) blow-up
4. Fine search: step through time to find time of closest approach and miss distance
5. Risk: collision probability from CDM covariance, or a transparent simplified estimate from TLEs
6. Explain: an LLM turns the geometry into 'why' plus a DODGE / WAIT / WATCH recommendation
7. Visualise: a CesiumJS / Three.js 3D globe with the conjunction, a timeline and a recommendation panel
8. Validate: run the screener on the TraCSS answer key and report accuracy

STEP-BY-STEP BUILD

Phase 1 - Data + globe	Fetch and cache TLEs; parse with Skyfield; render a handful of satellites on a 3D globe.
Phase 2 - Closest approach	Vector subtraction over a time grid to find time of closest approach and miss distance for a pair.
Phase 3 - Screening at scale	Coarse pre-filters to shortlist candidate pairs, then fine search; validate on the TraCSS spherical-volume answer key.
Phase 4 - Risk + explanation	Simplified collision probability first, then CDM-covariance probability where available; LLM explanation + DODGE / WAIT / WATCH.
Phase 5 - 3D UX	Highlight conjunctions, add a timeline and a recommendation panel on the globe.
Phase 6 - Accuracy (+ stretch)	Report recall vs the TraCSS key; stretch: live CDM ingestion from Space-Track.

REALISTIC RESULT YOU CAN CLAIM Headline you can defend: 'our screener reproduces the official TraCSS answer-key conjunctions (spherical volume) at X percent recall, with a live 3D globe from public TLEs and an explainable DODGE / WAIT call.' The numbers are credible because they are checked against an OFFICIAL key, not a toy.

WHAT CAN BREAK / WHAT WE ARE MISSING

- TLEs carry NO covariance, so a proper collision probability needs CDM covariance - do not overclaim probability from TLEs alone; use CDMs or a clearly-labelled simplified estimate.
- SGP4 / TLE accuracy is kilometre-level and degrades with propagation time, so screening is approximate (the TraCSS key lets you quantify this honestly).
- All-pairs screening is O(N squared) per time step - without coarse pre-filters it will not scale; the sieve IS the real engineering.
- Reference frames and time systems (TEME to ECI / ECEF, UTC vs TT) are easy to get wrong and silently corrupt results.
- NORAD 5-digit catalog numbers run out around 12 July 2026; new objects get 6-digit IDs with NO TLE-format data - use OMM / JSON.
- TraCSS files are by-request on Google Drive - request access EARLY; CelesTrak rate limits mean you must cache.
- SpaceX Stargaze and commercial SSA already screen conjunctions - your wedge is explainability + small / Indian operators, not raw screening.

RATINGS

Resume / recruiter pull	4/5	Real-world impact	3/5
Technical depth	4/5	Originality	4/5
Demo legibility	5/5	Buildability	3/5

PROS

- Live free data AND an official answer key - real, defensible numbers, not a toy globe.
- Huge judge wow (a live 3D globe of real orbits with collision alerts).
- Standout, hard-to-fake resume line (aerospace + ML + visualisation).
- 100 percent software and public data; strong India angle (Pixxel / GalaxEye / ISRO).

CONS / RISKS

- Niche user base and unclear monetisation versus the exam ideas.
- Orbital-mechanics correctness is unforgiving - the math must be right under demo pressure.
- A defensible collision probability needs covariance (CDMs), adding plumbing.

MVP - lock this demo path One primary satellite vs a curated catalog subset: screen + closest approach + simplified risk + 3D globe + TraCSS spherical-volume validation.

STRETCH Full all-vs-all screening with pre-filters, CDM-covariance probability, and a live Space-Track CDM feed.

TEAM (5 lanes) Data + propagation (Skyfield / SGP4) · Screening algorithm + TraCSS validation · Risk / probability + LLM explainer · 3D globe (CesiumJS / Three.js) · Backend / API + integration

STACK Python (Skyfield / SGP4, NumPy); CelesTrak GP + Space-Track APIs; TraCSS verification dataset; claude-opus-4-8 for explanations; CesiumJS / Three.js + Next.js.

OrbitGuard — scores at a glance

Axis	OrbitGuard
Data ready now	4
Demo wow	5
Honest metric	4
Low build risk	3
Novelty	4
Sellable / product	2
Recruiter pull	4
Overall	**8.2**

CHOSEN PROJECT: OrbitGuard — Space & Aerospace

Maximum judge wow (live 3D globe of real orbits + collision alerts). Standout resume line that is hard to fake (aerospace + ML + visualisation). 100 percent software + 100 percent public data. Strong India angle (Pixxel / GalaxEye / ISRO). The TraCSS answer key (Jan 2026) gives a real, defensible recall metric — not a toy globe.

LOCK THESE ON DAY ZERO — DO THESE FIRST

- **1.** REQUEST TraCSS dataset access NOW (Google account at the Office of Space Commerce portal) — files are on Google Drive and access is not instant.
- **2.** Create a free Space-Track.org account to get CDM covariance data.
- **3.** Cache a CelesTrak GP TLE/OMM snapshot locally — do NOT hammer the API (100 MB+/day; IP firewall risk).
- **4.** Choose ONE primary satellite to anchor the MVP (e.g. a Pixxel or GalaxEye satellite, or ISRO TechSat).
- **5.** Confirm your team's five lanes: propagation / screening+validation / risk+LLM / 3D globe / backend.
- **6.** Use OMM / JSON format for new objects (NORAD 5-digit IDs run out ~12 July 2026).

Key sources (June 2026): CelesTrak GP (TLE / OMM, free, 2-hour refresh); Space-Track.org (CDMs, free account); Skyfield (Python SGP4 propagator); TraCSS 'Dataset for Conjunction Assessment Verification' (Office of Space Commerce, Jan 2026, spherical + SFSH volume answer keys); Cesium ion / CesiumJS (3D globe, free tier); Three.js; Next.js; claude-opus-4-8 (LLM explainer).